

6502 NOPper

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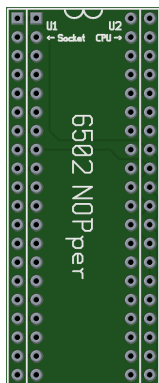
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While glossing over some ancient Commodore documentation I came across a simple debugging device which I have christened the 6502 NOPper. It contains exactly one IC: a 6502 CPU of which the data bus is hardwired to a hex value of \$EA or the opcode for NOP.

The purpose of this setup is to aide in finding address bus and decoding faults by temporarily replacing the systems 6502 CPU with the 6502 NOPper and switching it on. The 6502 receives all signals from the system, apart from the data bus. The net result is that after power on the CPU reads the reset vector (which is \$EA EA) and starts executing from that address where it reads \$EA or NOP. It then advances the program counter, only to find another NOP at the next address. The net result is that the address bus continuously increments between \$0000 and \$FFFF and all address lines carry square waves. Using an oscilloscope, it is now fairly easy to check the address bus, as each line should carry a square wave of a known frequency. As all addresses are looped through continuously, all address decoders of the system should become active at one time or another.

The practical version

A small PCB has been designed to make using the 6502 NOPper somewhat easier. Here it is:

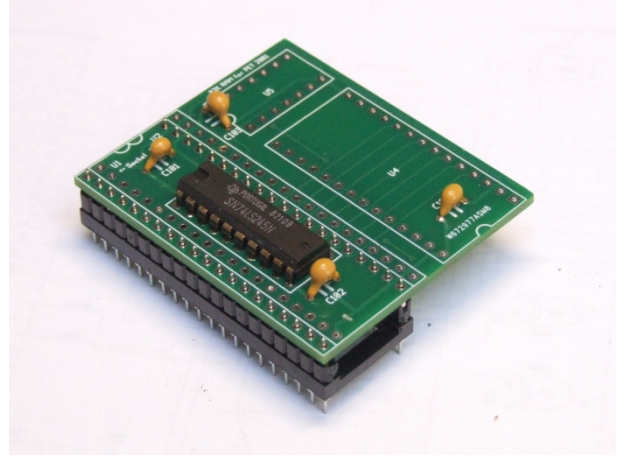
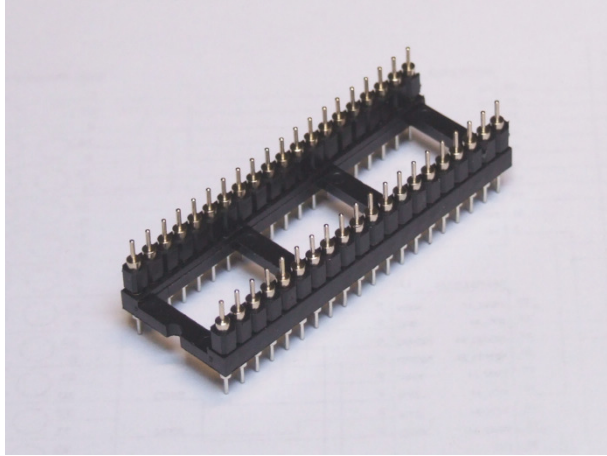


On the bottom side, there are 40 pins fitting an IC socket of a 6502, on top is an IC socket. In many 6502 systems the CPU is already socketed. By removing it and putting it in the 6502 NOPper, and then inserting the 6502 NOPper in the CPU socket of the system and powering it on, the test can be executed.

Building the PCB

The 6502NOPper is built the easiest in the following order:

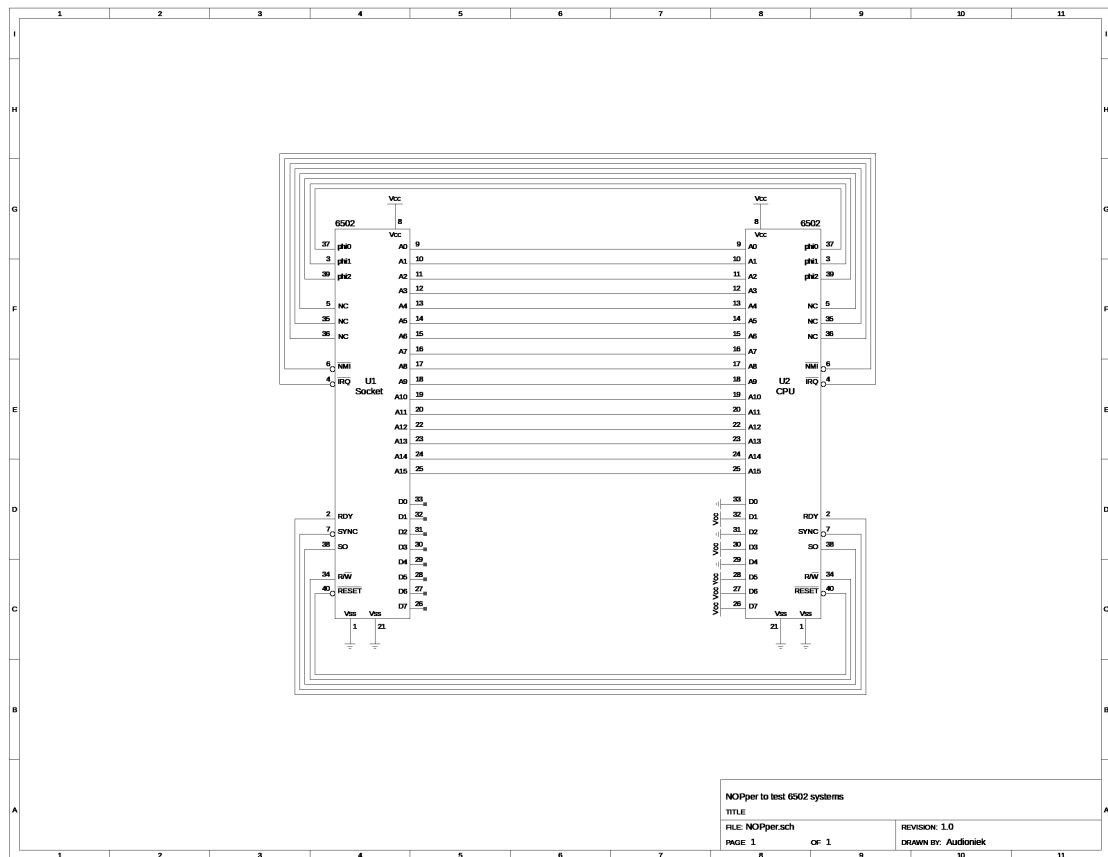
- Use the socket for the CPU as a template by inserting the thin pins of the pin strips into it and insert the thick end of the pins into the PCB holes intended for them.
(Note: different PCB shown, but you will get the idea.);



- Solder the thick ends of the strips on the component side of the PCB.
- Carefully remove the socket for the EPROM from the pin strips;
- Finish the 6502NOPper by soldering the socket for the CPU on the solder side of the PCB.

Schematic diagram

Here is the schematic diagram.



Appendix A: frequencies of the signals on the address lines

As the CPU is continuously executing NOPs and this instruction requires two clock cycles, the resulting frequencies on the various address lines should be:

Axx	1 MHz clock	2 MHz clock	3 MHz clock
A15	500 kHz	1 MHz	1.5 MHz
A14	250 kHz	500 kHz	750 kHz
A13	125 kHz	250 kHz	375 kHz
A12	62.5 kHz	125 kHz	187.5 kHz
A11	31.25 kHz	62.5 kHz	93.75 kHz
A10	15.625 kHz	31.25 kHz	46.875 kHz
A9	7812.5 Hz	15.625 kHz	23.4375 kHz
A8	3906.25 Hz	7812.5 Hz	11.71875 kHz
A7	1953/125 Hz	3906.25 Hz	5859.375 Hz
A6	976.5625 Hz	1953/125 Hz	2929.6875 Hz
A5	488.28125 Hz	976.5625 Hz	1464.84375 Hz
A4	244.140625 Hz	488.28125 Hz	732.421875 Hz
A3	122.0703125 Hz	244.140625 Hz	366.2109375 Hz
A2	61.03515625 Hz	122.0703125 Hz	183.10546875 Hz
A1	30.517578125 Hz	61.03515625 Hz	91.552734375 Hz
A0	15.2587890625 Hz	30.517578125 Hz	45.7763671875 Hz

Appendix B: cycle times of the signals on the address lines

As the CPU is continuously executing NOPs and this instruction requires two clock cycles, the resulting cycles times on the various address lines should be:

Axx	1 MHz clock	2 MHz clock	3 MHz clock
A15	2 μ s	1 μ s	0.75 μ s
A14	4 μ s	2 μ s	1.5 μ s
A13	8 μ s	4 μ s	3 μ s
A12	16 μ s	8 μ s	6 μ s
A11	32 μ s	16 μ s	12 μ s
A10	64 μ s	32 μ s	24 μ s
A9	128 μ s	64 μ s	48 μ s
A8	256 μ s	128 μ s	96 μ s
A7	512 μ s	256 μ s	192 μ s
A6	1.024 ms	512 μ s	384 μ s
A5	2.048 ms	1.024 ms	768 μ s
A4	4.096 ms	2.048 ms	1.536 ms
A3	8.192 ms	4.096 ms	3.072 ms
A2	16.384 ms	8.192 ms	6.144 ms
A1	32.768 ms	16.384 ms	12.288 ms
A0	65.536 ms	32.768 ms	24.576 ms

Appendix C: Bill of materials

Ref.	Description	Value	Qty.	Remark
U1	Pin strip, 20 pins	-	2	Use two strips of 20 pins if you can get them, otherwise use one 40 pin strip and cut it in half.
U2	IC socket, 40 pin, 600 mil wide	-	1	
-	PCB		1	